

Supplementary Information

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Supplementary Table S2	Backbone-superposition RMSD of FcμR-D1 alanine mutants onto wild-type chain C (PDB 7YTE)
Supplementary Data 1	FoldX AlaScan per-residue $\Delta\Delta G$ outputs and run commands for all seven FcμR-Fcμ complexes
Source Data	CSV files providing the numerical data behind Fig. 2-4 and the multiple sequence alignment (FASTA) underlying the evolutionary conservation analysis in Fig. 7

Description of Supplementary Files

Supplementary Table S1. Complete list of FcμR-D1 residues within 5 Å of IgM heavy-chain atoms in each complex, together with interaction types assigned by PLIP and PDBePISA. Contact residue ranges are reported per chain.

Supplementary Table S2. Superposition results of all alanine-mutant models onto the wild-type FcμR-D1 chain C of 7YTE. The table reports the total number of aligned atoms before and after outlier rejection. The RMSD of 0.000 Å for the refined atom pairs reflects the exclusion of mutated side-chain atoms by the alignment algorithm, leaving only identical backbone atoms.

Supplementary Data 1. Complete FoldX AlaScan per-residue $\Delta\Delta G$ values for all seven complexes, together with the command-line parameters used for every run.

Source Data. CSV files providing the numerical data behind Fig. 2-4 and the multiple sequence alignment (FASTA) underlying the evolutionary conservation analysis in Fig. 7.

Additional Notes

All supplementary tables are provided as separate, editable Excel files (.xlsx). No supplementary figures are included. Software and code availability information is provided in the main article.